# ON LOCALIZATION PHENOMENA IN PLASTICITY MODELS

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Abstract.

Contents



## 1. Introduction and notation

## 2. The tensile test at small deformations

In this section we develop a one dimensional model for an elastoplastic bar at small deformations. We prove the existence of solutions and we study the boundary value problem that describe tensile test. We show that the model is not able to predict the onset of necking.

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2.1. **Multidimensional kinematics.** Let $\Omega = (0, L) \times \Sigma$ with $L > 0$ and $\Sigma \subset \mathbb{R}^{d-1}$ be the reference configuration of the cylindrical bar. We denote the generic point in $\Omega$ as $(x, y)$ with $x \in (0, L)$ and $y \in \Sigma$. We assume a macroscopic displacement of the form

$$\begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} u_1(x) \\ u_2(x)y \end{pmatrix}$$

where $u_1$ and $u_2$ are functions from $(0, L)$ to $\mathbb{R}$. This assumption means that the sections of the cylindrical bar remain flat and can translate and expand uniformly. The corresponding displacement gradient is

$$H = \begin{pmatrix} u_1' & 0 \\ u_2' y & u_2 I \end{pmatrix}.$$

The plastic distortion is assumed to be of the form

$$H_p = \begin{pmatrix} p & 0 \\ 0 & -\frac{p}{d-1} I \end{pmatrix}$$

where $p\colon (0, L) \to \mathbb{R}$. This choice implies plastic incompressibility. The elastic distortion is then given by

$$H_e = H - H_p = \begin{pmatrix} u_1' - p & 0 \\ u_2' y & \left(u_2 + \frac{p}{d-1}\right) I \end{pmatrix}.$$

Then the elastic strain is

$$E_e = \begin{pmatrix} u_1' - p & \frac{u_2'}{2} y^T \\ \frac{u_2'}{2} y & \left(u_2 + \frac{p}{d-1}\right) I \end{pmatrix} = \begin{pmatrix} e_1 & \frac{u_2'}{2} y^T \\ \frac{u_2'}{2} y & e_2 I \end{pmatrix}$$

where we have introduced $e_1 = u_1' - p$ and $e_2 = u_2 + (d-1)^{-1} p$. Assuming an isotropic elastic response the elastic free energy density is given by

$$\psi_e = \frac{1}{2}\kappa (\operatorname{tr} E_e)^2 + \mu |\operatorname{dev} E_e|^2$$

where $\kappa \geq 0$ and $\mu \geq 0$ are the bulk and shear modulus respectively. We notice that

$$\operatorname{tr} E_e = e_1 + (d-1) e_2 \quad \text{and} \quad \operatorname{dev} E_e = \begin{pmatrix} \frac{d-1}{d}(e_1 - e_2) & \frac{u_2'}{2} y^T \\ \frac{u_2'}{2} y & -\frac{1}{d}(e_1 - e_2) I \end{pmatrix}.$$

Then the elastic free energy density can be written as

$$\psi_e = \frac{1}{2}\kappa \left(e_1 + (d-1) e_2\right)^2 + \frac{d-1}{d}\mu (e_1 - e_2)^2 + \frac{1}{2}|y|^2 \mu |u_2'|^2.$$

We conclude that the elastic free energy per unit axial length is

$$\int_\Sigma \psi_e dy = \frac{1}{2} A\kappa \left(e_1 + (d-1) e_2\right)^2 + \frac{d-1}{d} A\mu (e_1 - e_2)^2 + \frac{1}{2} J\mu |u_2'|^2$$

where

$$A = \int_\Sigma dy \quad \text{and} \quad J = \int_\Sigma |y|^2 dy$$

are the polar and the polar second order moment of area.

2.2. **One dimensional kinematics.**

## 3. The tensile test at large deformations

## 4. Shear band localization in cohesive materials

## Appendix A. Modeling in continuum mechanics

## Appendix B. Tools from functional analysis

### B.1. **Other results.**

**Proposition B.1.** (Gronwall inequality) Let $z \in L^\infty((0,T))$. Let $C \in \mathbb{R}$ and $\alpha, \beta$ in $L^1((0,T),[0,\infty))$ such that

$$y(t) \le C + \int_0^t (\alpha(s)y(s) + \beta(s))\,ds \quad \text{for a.e. } t \in (0,T).$$

Then

$$z(t) \le e^{\int_0^t \alpha(s)ds} \left( C + \int_0^t \beta(s) e^{-\int_0^s \alpha(r)dr} \right) \quad \text{for a.e. } t \in (0,T).$$

*Proof.* Let

$$Z(t) = C + \int_0^t (\alpha(s)z(s) + \beta(s))\,ds$$

so that $z(t) \le Z(t)$. Clearly $Z \in \mathrm{AC}([0,T])$. It holds that

$$\dot{Z}(t) = \alpha(t)z(t) + \beta(t) \le \alpha Z(t) + \beta(t) \quad \text{for a.e. } t \in (0,T).$$

Multiplying by $e^{-\int_0^t \alpha(s)ds}$ one gets

$$\frac{d}{dt}\left( Z(t) e^{-\int_0^t \alpha(s)ds} \right) \le \beta(t) e^{-\int_0^t \alpha(s)ds}$$

and integrating on $(0,t)$

$$Z(t) e^{-\int_0^t \alpha(s)ds} \le C + \int_0^t \beta(s) e^{-\int_0^s \alpha(r)dr} ds$$

Then

$$z(t) \le Z(t) \le e^{\int_0^t \alpha(s)ds} \left( C + \int_0^t \beta(s) e^{-\int_0^s \alpha(r)dr} \right) \quad \text{for a.e. } t \in (0,T).$$

□

### B.2. **Nonlinear analysis.**

**Proposition B.2.** Let $V$ a Banach space and let $f \in C^1(V)$ and $K \subset V$ compact. Then $f$ is Lipschitz continuous on $K$.

*Proof.* By contradiction let $u_n$ and $v_n$ two sequences in $K$ such that

$$\frac{|F(u_n) - F(v_n)|}{\|u_n - v_n\|} \ge n.$$

By compactness of $K$ up to extracting subsequences we can assume that $u_n \to u$ and $v_n \to v$. If $u \ne v$ then

$$\frac{|F(u_n) - F(v_n)|}{\|u_n - v_n\|} \to \frac{|F(u) - F(v)|}{\|u - v\|} < \infty$$

which is a contradiction. If $u = v$ then

$$\frac{|F(u_n) - F(v_n)|}{\|u_n - v_n\|} = \frac{|\langle DF(u), u_n - v_n\rangle + o(\|u_n - v_n\|)|}{\|u_n - v_n\|} \le \|DF(u)\| + o(1) \to \|DF(u)\| < \infty$$

which is a contradiction. $\square$

B.3. **Convex analysis.**

**Example B.3.** Let $f(v) = \frac{1}{2}\|v\|^2$. Then $f^*(\xi) = \frac{1}{2}\|\xi\|^2$. Indeed

$$\langle \xi, v\rangle - \frac{1}{2}\|v\|^2 \le \frac{2\|\xi\|\|v\| - \|v\|^2}{2} \le \frac{\|\xi\|^2}{2}.$$

This implies $f^*(\xi) \le \frac{\|\xi\|^2}{2}$. To get the opposite inequality, fix $\epsilon > 0$ and let $v \in V$ such that $\|v\| = \|\xi\|$ and $\langle \xi, v\rangle \ge \|\xi\|^2 - \epsilon$. Then

$$f^*(\xi) \ge \|\xi\|^2 - \epsilon - \frac{1}{2}\|\xi\|^2 \ge \frac{1}{2}\|\xi\|^2 - \epsilon.$$

By arbitrariness of $\epsilon$ we conclude.

Let $f$ and $g$ from $V$ to $(-\infty, \infty]$. Their inf-convolution $f \Box g \colon V \to (-\infty, \infty]$ is

$$f \Box g(v) = \inf_{w \in V} (f(w) + g(v - w)).$$

**Proposition B.4.** (inf-convolution duality formula) Let $f$ and $g$ from $V$ to $(-\infty, \infty]$. Then

(i) $(f \Box g)^* = f^* + g^*$
(ii) $(f + g)^* \le f^* \Box g^*$
(iii) if $f$ and $g$ are convex and proper and there exists $v \in \operatorname{dom} f \cap \operatorname{dom} g$ and $f$ is continuous at $v$ then $(f + g)^* = f^* \Box g^*$.

*Proof. Step 1.* It holds that

$$\begin{aligned}
(f \Box g)^* &= \sup_{v \in V} \left( \langle \xi, v\rangle - \inf_{w \in V} (f(w) + g(v - w)) \right) = \\
&= \sup_{v \in V} \sup_{w \in V} (\langle \xi, w\rangle - f(w) + \langle \xi, v - w\rangle - g(v - w)) = \\
&= \sup_{w \in V} \sup_{v \in V} (\langle \xi, w\rangle - f(w) + \langle \xi, v - w\rangle - g(v - w)) = \\
&= \sup_{w \in V} (\langle \xi, w\rangle - f(w) + g^*(\xi)) = f^*(\xi) + g^*(\xi).
\end{aligned}$$

This establish (i).

*Step 2.* By the Young–Fenchel inequality

$$f^*(\zeta) + g^*(\xi - \zeta) \ge \langle \xi, v\rangle - f(v) - g(v).$$

Taking the supremum over $v \in V$

$$f^*(\zeta) + g^*(\xi - \zeta) \ge (f + g)^*$$

and taking the infimum over $\zeta \in V$

$$f^* \Box g^* \ge (f + g)^*.$$

This proves (ii).

*Step 3.* Now we prove (iii). If $(f+g)^*(\xi)=\infty$ then (iii) is satisfied at $\xi$. Then suppose that $(f+g)^*(\xi)\in[-\infty,\infty)$. Since $\operatorname{dom} f\cap\operatorname{dom} g\neq\emptyset$ then actually $(f+g)^*(\xi)=c\in(-\infty,\infty)$. Let

$$A=\{(a,v)\in\mathbb{R}\times V \mid a\le\langle\xi,v\rangle-g(v)-c\}.$$

Clearly $A$ is nonempty and convex. We show that $A$ and $\operatorname{int}(\operatorname{epi} f)$ are disjoint. Indeed if by contradiction $(a,v)\in A\cap\operatorname{int}(\operatorname{epi} f)$ then

$$f(v)<a\le\langle\xi,v\rangle-g(v)-c$$

and

$$c<\langle\xi,v\rangle-f(v)-g(v)\le(f+g)^*(\xi)=c.$$

Then by Hahn–Banach theorem there exists $(\gamma,\zeta)\in\mathbb{R}\times V^*$ such that

$$\gamma a+\langle\zeta,v\rangle\le\gamma b+\langle\zeta,w\rangle \tag{B.1}$$

for all $(a,v)\in\operatorname{int}(\operatorname{epi} f)$ and $(b,w)\in A$. If we fix $v=w\in\operatorname{dom} f\cap\operatorname{dom} g$, $b\in\mathbb{R}$ sufficiently small such that $(b,v)\in A$ and we let $a\to\infty$, we discover that $\gamma\le 0$ necessarily. If $\gamma=0$ then inequality (B.1) and Hahn–Banach theorem imply that $\operatorname{dom} f$ and $\operatorname{dom} g$ are separated by a closed hyperplane. This in turn implies that $\operatorname{int}(\operatorname{dom} f)$ and $\operatorname{dom} g$ are disjoint, but this contradicts the hypotheses so that $\beta<0$. In this case we set $\theta=|\beta|^{-1}f\zeta$ and from inequality (B.1) we get

$$\begin{aligned}
f^*(\theta)&=\sup\{\langle\theta,v\rangle-f(v)\mid v\in V\}=\\
&=\sup\{\langle\theta,v\rangle-a\mid (a,v)\in\operatorname{epi} f\}\le\\
&\le\inf\{\langle\theta,v\rangle-a\mid (a,v)\in A\}=\\
&=c+\inf\{\langle\theta-\xi,v\rangle+g(v)\mid v\in V\}=c-g^*(\xi-\theta).
\end{aligned}$$

Then

$$f^*\Box g^*(\xi)\le f^*(\theta)+g^*(\xi-\theta)\le c=(f+g)^*(\xi).$$

This concludes the proof. □

**Proposition B.5** (properties of positively 1-homogeneous convex functions)**.** Let $f\colon V\to(-\infty,\infty]$ be a proper, l.s.c., positively 1-homogeneous and convex function. Then

(i) $f(0)=0$ and $f$ is subadditive
(ii) for each $x\in\operatorname{dom} f$ it holds $\partial f(x)=\{\xi\in\partial f(0)\mid f(x)=\langle\xi,x\rangle\}$
(iii) $f^*=\iota_K$ with $K=\partial f(0)$ and $f=\sigma_K$.

*Proof. Step 1.* Let $v\in\operatorname{dom} f$. Since $f$ is l.s.c. and is positive 1-homogeneous

$$f(0)\le\liminf_{n\to\infty} f\left(\frac{1}{n}v\right)=\liminf_{n\to\infty}\frac{f(v)}{n}=0.$$

By positively 1-homogeneity and convexity we have

$$\frac{1}{2}f(v_0+v_1)=f\left(\frac{v_0+v_1}{2}\right)\le\frac{f(v_0)+f(v_1)}{2}.$$

This proves (i).

*Step 2.* We prove that for all $\xi\in V^*$ it holds that $f^*(\xi)\in\{0,\infty\}$. Indeed let $\xi\in V^*$ such that $f^*(\xi)<\infty$. Then since $f$ is positively 1-homogeneous, it holds $f^*(\xi)=2^{-1}f^*(\xi)$ and this implies $f^*(\xi)=0$.

*Step 3.* Let $v \in \operatorname{dom} f$ and $\xi \in \partial f(v)$. This is equivalent to the equality $f(v) + f^*(\xi) = \langle \xi, v\rangle$ which in turn implies that $f^*(\xi) < \infty$ and by previous step $f^*(\xi) = 0$. Then $f(v) = \langle \xi, v\rangle$ and for all $w \in V$

$$f(w) \geq f(v) + \langle \xi, w - v\rangle = \langle \xi, w\rangle.$$

This is equivalent to $\xi \in \partial f(0)$. We have proved that if $\xi \in \partial f(v)$ then $f(v) = \langle \xi, v\rangle$ and $v \in \partial f(0)$. It is immediate to check that also the converse is true. This proves (ii).

*Step 4.* It holds that $\xi \in K$ if and only if $f^*(\xi) = f(0) + f^*(\xi) = \langle \xi, 0\rangle = 0$. This proves that $f^* = \iota_K$. Since $f$ is l.s.c. then $f = f^* * = \iota_K^* = \sigma_K$. This establishes (iii). $\square$

B.4. **Bochner spaces.**

**Definition B.6.** Let $V$ be a Banach space. We say that $u\colon [0,T] \to V$ is absolutely continuous and we write $u \in AC([0,T],V)$ if there exists $m \in L^1((0,T),(0,\infty))$ such that

$$\|u(t) - u(s)\| \leq \int_0^T m(r)dr \quad \text{for all } 0 \leq s < t \leq T.$$

Given $u\colon [0,T] \to V$ and $0 \leq a < b \leq T$ we define the total variation on $[a,b]$

$$\operatorname{Var}(u,[a,b]) = \sup\left\{ \sum_{k=1}^N \|u(t_k) - u(t_{k-1})\| \;\middle|\; a = t_0 < \cdots < t_N = b \right\}.$$

We say that $u$ has bounded variation and we write $u \in BV([0,T],V)$ if $\operatorname{Var}(u,[0,T]) < \infty$.

**Proposition B.7.** Let $V$ be a Banach space. Then $W^{1,1}((0,T),V) \subset AC([0,T],V)$. If $V$ is separable and reflexive the equality holds.

*Proof.* Let $u \in W^{1,1}((0,T),V)$. Then

$$\|u(t) - u(s)\| = \left\| \int_s^t \dot u(r)dr \right\| \leq \int_s^t \|\dot u(r)\| dr.$$

This proves that $u \in AC([0,T],V)$ with $m = \|\dot u\|$.

Now suppose that $V$ is reflexive. We have that

$$\left\| \frac{u(t+h) - u(t)}{h} \right\| \leq \frac{1}{h}\int_t^{t+h} m(r)dr.$$

By the Lebesgue differentiation theorem the left hand side is bounded as $h \to 0$ for a.e. $t \in (0,T)$. Then for each $h_n \to 0$, up to extracting a subsequence,

$$\frac{u(t+h_n) - u(t)}{h_n} \rightharpoonup v \quad \text{for some } v \in V.$$

We prove that $v$ is common to each sequence. Since $V^*$ is separable and reflexive then there exists $\xi_i$ dense in $V^*$. Now for each $i$ the map $t \mapsto \langle \xi_i, u(t)\rangle$ is $AC([0,T])$. Then for a.e. $t \in (0,T)$

$$\lim_{h\to 0} \left\langle \xi_i, \frac{u(t+h) - u(t)}{h} \right\rangle \quad \text{exists for all } i. \tag{B.2}$$

Then if $h_n \to 0$ and $k_n \to 0$ are such that

$$\frac{u(t+h_n) - u(t)}{h_n} \rightharpoonup v \quad \text{and} \quad \frac{u(t+k_n) - u(t)}{k_n} \rightharpoonup w$$

from (B.2) it follows that $\langle \xi_i, v - w\rangle = 0$ for all $i$. This implies $v = w$. Up to now we have proved that there exists $v\colon (0,T) \to V$ such that

$$\frac{u(t+h) - u(t)}{h} \rightharpoonup v(t) \quad \text{for a.e. } t \in (0,T).$$

**Now one should prove that this convergence is actually strong. This will show automatically that $u \in W^{1,1}((0,T),V)$.** □

## Appendix C. Balanced-viscosity solutions

C.1. **Introduction.** Let

$$(V) \qquad V \text{ a separable and reflexive Banach space.}$$

Consider an evolution $u\colon [0,T] \to V$ and the doubly nonlinear equation for the evolution

$$\partial_u \mathcal{E}(u(t),t) + \partial\Phi(\dot u(t)) \ni 0 \quad \text{for } t \in (0,T). \tag{C.1}$$

Here $\mathcal{E}$ is the energy and $\Phi$ is the dissipation potential. In application usually $\mathcal{E}(u,t) = \Psi(u) - \langle \mathcal{F}(t), u\rangle$ where $\Psi$ is the free energy and $\mathcal{F}$ is the external action. For a given $\eta > 0$ consider the time-rescaled evolution $u\colon (0, \eta^{-1}T) \to V$ and the time-rescaled equation

$$\partial_u \mathcal{E}(u(t),\eta t) + \partial\Phi(\dot u(t)) \ni 0 \quad \text{for } t \in (0,\eta^{-1}T). \tag{C.2}$$

For $\eta \to 0$ this describe the situation in which the external actions are mush slower with respect to the characteristic relaxation time of the system. Equation (C.2) can be written as

$$\partial_u \mathcal{E}(u_\eta(t),t) + \partial\Phi_\eta(\dot u_\eta(t)) \ni 0 \quad \text{for } t \in (0,T) \tag{C.3}$$

where $u_\eta(t) = u(\eta^{-1}t)$ and $\Phi_\eta(v) = \eta^{-1}\Phi(\eta v)$.

C.2. **Assumptions.** The energy $\mathcal{E}$ is supposed to satisfy the following assumptions.

($\mathcal{E}$.1) $\quad \mathcal{E} \in C^1(V \times [0,T])$

($\mathcal{E}$.2) $\quad$ for all $E > 0$ the sublevel $D_E = \{u \in V \mid \sup_{t\in[0,T]} \mathcal{E}(u,t) \le D_E\}$ is weakly compact

($\mathcal{E}$.3) $\quad$ there exists $C > 0$ such that $|\partial_t \mathcal{E}(u,t)| \le C(1 + \mathcal{E}(u,t))$ for all $u \in V$ and $t \in [0,T]$

The dissipation potential is defined by

$$\Phi(v) = \mathcal{R}(v) + \mathcal{V}(v)$$

and for $\eta > 0$ its rescaled version is

$$\Phi_\eta(v) = \eta^{-1}\Phi_1(\eta v) = \mathcal{R}(v) + \eta^{-1}\mathcal{V}(\eta v) = \mathcal{R}(v) + \mathcal{V}_\eta(v).$$

Here $\mathcal{R}$ is a rate-independent dissipation potential and $\mathcal{V}$ is a rate-dependent dissipation potential that models viscosity. It is assumed that $\mathcal{R}$ and $\mathcal{V}$ satisfy the following assumptions.

($\Phi$.1) $\quad \mathcal{R}$ and $\mathcal{V}$ from $V$ into $[0,\infty)$ are continuous, convex and strictly positive in $V \setminus \{0\}$

($\Phi$.2) $\quad \mathcal{R}$ is positively 1-homogeneous

Notice that from Proposition B.5 it holds $\mathcal{R}(0) = 0$.

($\Phi$.3) $\quad \mathcal{V}(v) = \phi(\|v\|)$ with $\phi \in C^1([0,\infty))$ convex such that $\phi(0) = 0$, $\phi'(0) = 0$, $\phi(r) > 0$ for $r > 0$ and $\lim_{r\to\infty} \phi'(r) = \infty$

Notice that $\phi$ can be extended to an even $C^1(\mathbb{R})$ function.

C.3. **Preliminary results regarding the dissipation potential.** Let $K = \partial\mathcal{R}(0)$. By Proposition B.5 it follows that $\mathcal{R} = \sigma_K$ and $\mathcal{R}^* = \iota_K$. Notice that there exists a constant $M > 0$ such that $\mathcal{R}(v) \leq M\|v\|$ for all $v \in V$. Indeed by contradiction assume that there exist $v_n \in V \setminus \{0\}$ such that $\mathcal{R}(v_n) \geq n\|v_n\|$. Since $\mathcal{R}$ is positively homogeneous the inequality implies $\mathcal{R}(n^{-1}v_n/\|v_n\|) \geq 1$ so $\liminf_{n\to\infty} \mathcal{R}(n^{-1}v_n/\|v_n\|) \geq 1$. But $n^{-1}v_n/\|v_n\| \to 0$ so by continuity of $\mathcal{R}$ it follows $\lim_{n\to\infty} \mathcal{R}(n^{-1}v_n/\|v_n\|) = 0$. This is the desired contradiction.

Since $\Phi_1$ is convex and $\Phi_1(0) = 0$ then for all $v \in V$ the functions $\eta \mapsto \Phi_\eta(v)$ are increasing. Then it is well defined

$$\Phi_0(v) = \lim_{\eta\to 0^+} \Phi_\eta(v) = \inf_{\eta>0} \Phi_\eta(v) = \mathcal{R}(v).$$

Last inequality holds true since $\phi(0) = 0$ and $\phi'(0) = 0$.

It simple to check that $\Phi_\eta^*(\xi) = \eta^{-1}\Phi^*(\xi)$. It is possible to obtain a more explicit representation formula for $\Phi^*$. By the inf-convolution duality formula

$$\Phi_\eta^*(\xi) = (\mathcal{R} + \mathcal{V}_\eta)^*(\xi) = \inf_{\zeta\in V} \left(\mathcal{R}^*(\zeta) + \mathcal{V}_\eta^*(\xi - \zeta)\right) = \eta^{-1} \inf_{\zeta\in K} \mathcal{V}^*(\xi-\zeta) = \eta^{-1} \min_{\zeta\in K} \mathcal{V}^*(\xi-\zeta). \tag{C.4}$$

Last equality holds true because, since $K$ is closed and convex, it is also weakly closed, $\mathcal{V}$ is convex and coercive, then the infimum is actually a minimum. Now $\mathcal{V}^*(\xi) = \phi^*(\|\xi\|)$.

**Lemma C.1.** *The function $\phi^*$ satisfies $\phi^*(0) = 0$, $(\phi^*)'(0) = 0$ and is monotone.*

*Proof.* Let $r = \rho(\gamma)$ the solution to $\gamma - \phi'(\rho(\gamma)) = 0$. Clearly $\rho \in C^1(\mathbb{R})$ and $\rho(0) = 0$. Then $\phi^*(\gamma) = \gamma\rho(\gamma) - \phi(\rho(\gamma))$. Now $\phi^*(0) = -\phi(0) = 0$ and $(\phi^*)'(0) = -\phi'(0) = 0$. □

Using that lemma and (C.4)

$$\Phi_\eta^*(\xi) = \eta^{-1} \min_{\zeta\in K} \phi^*(\|\xi - \zeta\|). \tag{C.5}$$

C.4. **Absolutely continuous and BV evolutions.** Let $\mathcal{R} \colon V \to [0,\infty)$ satisfying ($\Phi$.1) and ($\Phi$.2). We say that a map $u \colon [0,T] \to V$ is $\mathrm{AC}_\mathcal{R}([0,T],V)$ if there exists $m \in L^1((0,T),[0,\infty))$ such that

$$\mathcal{R}(u(t_1) - u(t_0)) \leq \int_{t_0}^{t_1} m(s)ds \quad \text{for every } 0 \leq t_0 < t_1 \leq T. \tag{C.6}$$

**Lemma C.2.** *Let $u \in AC_\mathcal{R}([0,T],V)$. Then*

$$\mathcal{R}[\dot{u}](t) \doteq \lim_{h\to 0} \mathcal{R}\left(\frac{u(t+h) - u(t)}{h}\right)$$

*exists for a.e. $t \in (0,T)$ and $\mathcal{R}[\dot{u}] \in L^1((0,T))$. Moreover for all $m$ as in* (C.6)

$$\mathcal{R}[\dot{u}](t) \leq m(t) \quad \text{for a.e. } t \in (0,T).$$

*Proof.* For any $s \in (0,T)$ and $t \in (0,T)$ let $\delta_s(t) = \mathcal{R}(u(t) - u(s))$. Then

$$\delta_s(t_1) - \delta_s(t_0) \leq \mathcal{R}(u(t_1) - u(t_0)) \leq \int_{t_0}^{t_1} m(s)ds \quad \text{for all } 0 \leq t_0 < t_1 \leq T.$$

The second inequality holds since $\mathcal{R}$ is sub additive by Proposition B.5 so

$$\delta_s(t_1) = \mathcal{R}(u(t_1) - u(s)) \leq \mathcal{R}(u(t_1) - u(t_0)) + \mathcal{R}(u(t_0) - u(s)) = \mathcal{R}(u(t_1) - u(t_0)) + \delta_s(t_0).$$

Then the map $t \mapsto \delta_s(t) - \int_0^t m(r)dr = \beta_s(t)$ is non increasing on $(0,T)$ and in particular it is differentiable a.e. on $(0,T)$. This in turns implies that $\delta_s$ is differentiable a.e. on $(0,T)$ and

$$\dot{\delta}_s(t) \leq \delta(t) \doteq \liminf_{h\to 0^+} \mathcal{R}\left(\frac{u(t+h)-u(t)}{h}\right) \quad \text{for a.e. } t \in (0,T). \tag{C.7}$$

Note that

$$0 \leq \delta(t) \leq \liminf_{h\to 0^+} \frac{1}{h}\int_t^{t+h} m(s)ds = m(t) \quad \text{for a.e. } t \in (0,T) \tag{C.8}$$

and this implies $\delta \in L^1((0,T))$. Since $\delta_s(t) = \beta_s(t) + \int_0^t m(r)dr$ and $\beta_s$ is non increasing then the singular part of the distributional derivative of $\delta_s$ is non positive and

$$\mathcal{R}(u(t)-u(s)) = \delta_s(t) \leq \int_s^t \dot{\delta}_s(r)dr \leq \int_s^t \delta(r)dr. \tag{C.9}$$

Let

$$\gamma(t) \doteq \limsup_{h\to 0^+} \mathcal{R}\left(\frac{u(t+h)-u(t)}{h}\right).$$

By (C.9) then $\delta(t) \leq \gamma(t) \leq \delta(t)$. This proves that

$$\delta^+(t) \doteq \lim_{h\to 0^+} \mathcal{R}\left(\frac{u(t+h)-u(t)}{h}\right) = \delta(t) \quad \text{exists for a.e. } t \in (0,T).$$

Applying the previous argument to $t \mapsto u(T-t)$ and $v \mapsto \mathcal{R}(-v)$ in place of $u$ and $\mathcal{R}$ respectively, follows that

$$\delta^-(t) = \lim_{h\to 0^-} \mathcal{R}\left(\frac{u(t+h)-u(t)}{h}\right) = \lim_{h\to 0^+} \mathcal{R}\left(\frac{u(t)-u(t-h)}{h}\right) \quad \text{exists for a.e. } t \in (0,T)$$

and satisfy the conditions

$$\mathcal{R}(u(t)-u(s)) \leq \int_s^t \delta^-(r)dr. \tag{C.10}$$

and $\delta^-(t) \leq m(t)$ for a.e. $t \in (0,T)$. This means that $\delta^+$ and $\delta^-$ coincides both with the minimal $m \in L^1((0,T),(0,\infty))$ for which (C.6) holds. □

Given a map $u\colon [0,T] \to V$ we define the total variation on $[a,b] \subset [0,T]$ as

$$\mathrm{Var}_{\mathcal{R}}(u,[a,b]) = \sup\left\{ \sum_{k=1}^N \mathcal{R}(u(t_k)) - \mathcal{R}(u(t_{k-1})) \;\middle|\; a = t_0 < \cdots < t_N = b \right\}.$$

We say that $u \in \mathrm{BV}_{\mathcal{R}}([0,T],V)$ if $\mathrm{Var}_{\mathcal{R}}(u,[0,T]) < \infty$.

**Lemma C.3.** *It holds that $AC([0,T],V) \subset AC_{\mathcal{R}}([0,T],V)$ and $BV([0,T],V) \subset BV_{\mathcal{R}}([0,T],V)$.*

*Proof.* Recall that there exists $M > 0$ such that $\mathcal{R}(v) \leq M\|v\|$. Let $u \in AC([0,T],V)$. Then

$$\mathcal{R}(u(t_1)-u(t_0)) \leq M\|u(t_1)-u(t_0)\| \leq \int_{t_0}^{t_1} Mm(s)ds.$$

This shows that $u \in \mathrm{AC}_{\mathcal{R}}([0,T],V)$. Next let $u \in \mathrm{BV}([0,T],V)$. Then

$$\mathrm{Var}_{\mathcal{R}}(u,[0,T]) \leq M\mathrm{Var}(u,[0,T]).$$

This implies that $u \in \mathrm{BV}_{\mathcal{R}}([0,T],V)$. □

C.5. **A chain rule formula for the energy functional.**

**Proposition C.1.** Let $u \in AC([0,T],V)$. Then $t \mapsto \mathcal{E}(u(t),t)$ is absolutely continuous in $[0,T]$ and

$$\frac{d}{dt}\mathcal{E}(u(t),t) = \langle \partial_u \mathcal{E}(u(t),t), \dot{u}(t)\rangle + \partial_t \mathcal{E}(u(t),t) \quad \text{for a.e. } t \in (0,T). \tag{C.11}$$

*Proof. Step 1.* Since $u$ is continuous then $K = u([0,T])$ is compact. Indeed let $u_n = u(t_n)$ be a sequence in $K$. Then up to a subsequence $t_n \to t$ and by continuity $u_n = u(t_n) \to u(t)$.

*Step 2.* Since $\mathcal{E} \in C^1(V \times [0,T])$ then by Proposition B.2 it is Lipschitz in $K \times [0,T]$. Let $L > 0$ be the Lipschitz constant.

*Step 3.* Let $0 \le s < t \le T$ and $m \in L^1((0,T),(0,\infty))$ as in Definition B.6. Then

$$|\mathcal{E}(u(t),t) - \mathcal{E}(u(s),s)| \le L\left(\|u(t)-u(s)\| + |t-s|\right) \le L \int_s^t (m(r)+r)\, dr.$$

This proves that $t \mapsto \mathcal{E}(u(t),t)$ is absolutely continuous in $[0,T]$.

*Step 4.* By Proposition B.7 it follows that $u$ is a.e. differentiable in $(0,T)$. This implies formula (C.11). □

C.6. **Existence results in presence of positive viscosity.**

**Proposition C.2.** Let $u_0 \in V$. Then there exists $u_\eta \in \mathrm{AC}([0,T],V)$ such that $u_\eta(0) = u_0$ and

$$\partial_u \mathcal{E}(u_\eta(t),t) + \partial\Phi_\eta(\dot{u}_\eta(t)) \ni 0 \quad \text{for a.e. } t \in (0,T).$$

Moreover the identity

$$\mathcal{E}(u_\eta(t),t) + \int_s^t \left(\Phi_\eta(\dot{u}_\eta(r)) + \Phi_\eta^*(-\partial_u\mathcal{E}(u_\eta(r),r)\right) dr = \mathcal{E}(u_\eta(s),s) + \int_s^t \partial_t\mathcal{E}(u_\eta(r),r)dr \tag{C.12}$$

holds for every $0 \le s < t \le T$.

*Proof.* I still have to report the proof of existence. We prove the identity (C.12). By the properties of subdifferentials

$$\Phi_\eta(\dot{u}_\eta(t)) + \Phi_\eta^*(-\partial_t\mathcal{E}(u_\eta(t),t)) = -\langle \partial_u\mathcal{E}(u_\eta(t),t), \dot{u}_\eta(t)\rangle \quad \text{for a.e. } t \in (0,T).$$

By (C.11) we have

$$\Phi_\eta(\dot{u}_\eta(t)) + \Phi_\eta^*(-\partial_t\mathcal{E}(u_\eta(t),t)) = -\frac{d}{dt}\mathcal{E}(u_\eta(t),t) + \partial_t\mathcal{E}(u_\eta(t),t) \quad \text{for a.e. } t \in (0,T).$$

Integrating this identity in $(s,t)$ we get the thesis. □

C.7. **Contact dissipation potential.** The dissipation contact potential $\mathfrak{p}\colon V \times V^* \to [0,\infty)$ is

$$\mathfrak{p}(v,\xi) = \inf_{\eta>0}\left(\Phi_\eta(v) + \Phi_\eta^*(\xi)\right) = \inf_{\eta>0} \eta^{-1}\left(\Phi(\eta v) + \Phi^*(\xi)\right).$$

By (C.5)

$$\begin{aligned}\mathfrak{p}(v,\xi) &= \inf_{\eta>0}\left(\mathcal{R}(v) + \eta^{-1}\phi(\eta\|v\|) + \eta^{-1}\min_{\zeta\in K}\phi^*(\|\xi-\zeta\|)\right) = \\ &= \mathcal{R}(v) + \inf_{\eta>0}\left[\eta^{-1}\min_{\zeta\in K}\left(\phi(\eta\|v\|) + \phi^*(\|\xi-\zeta\|)\right)\right] = \\ &= \mathcal{R}(v) + \min_{\zeta\in K}\inf_{\eta>0}\frac{\phi(\eta\|v\|)+\phi^*(\|\xi-\zeta\|)}{\eta}\end{aligned}$$

where last equality holds because the minimum point does not depends on $\eta$. But for all $r$ and $\gamma$ in $\mathbb{R}$

$$\inf_{\eta>0} \frac{\phi(\eta r) + \phi^*(\gamma)}{\eta} = \gamma r.$$

Indeed by Young–Fenchel inequality

$$\frac{\phi(\eta r) + \phi^*(\gamma)}{\eta} \geq \gamma r$$

and the identity holds for $\eta$ such that $\gamma = \phi'(\eta r)$. Then

$$\mathfrak{p}(v, \xi) = \mathcal{R}(v) + \|v\| \min_{\zeta \in K} \|\xi - \zeta\|. \tag{C.13}$$

Let $\mathfrak{f}\colon V \times V \times [0, T] \to [0, \infty)$ defined by

$$\mathfrak{f}(u, v, t) = \mathfrak{p}(v, -\partial_u \mathcal{E}(u, t)) = \mathcal{R}(v) + \mathfrak{e}(u, t)\|v\| \tag{C.14}$$

where $\mathfrak{e}\colon V \times [0, T] \to [0, \infty)$ is defined by

$$\mathfrak{e}(u, t) = \min_{\zeta \in K} \|\partial_u \mathcal{E}(u, t) + \zeta\|.$$

Notice that $\mathfrak{e}(u, t) = 0$ if and only if $-\partial_u \mathcal{E}(u, t) \in K$.

C.8. **Parametrized balanced-viscosity solutions.** Let $u_\eta \in \mathrm{AC}([0, T], V)$ be the solution to the doubly nonlinear equation

$$\partial_u \mathcal{E}(u_\eta(t), t) + \partial \Phi_\eta(\dot{u}_\eta(t)) \ni 0 \quad \text{for } t \in (0, T)$$

The identity (C.12) implies

$$\mathcal{E}(u_\eta(t), t) + \int_s^t \mathfrak{f}(u_\eta(t), \dot{u}_\eta(t), t) dr \leq \mathcal{E}(u_\eta(s), s) + \int_s^t \partial_t \mathcal{E}(u_\eta(t), t) dr \tag{C.15}$$

for all $0 \leq s < t \leq T$.

**Lemma C.4.** *There exists a constant $C > 0$ such that*

$$\int_0^T \mathfrak{f}(u_\eta(t), \dot{u}_\eta(t), t) dt \leq C \quad \textit{for all } \eta > 0.$$

*Proof.* Since $\mathfrak{f} \geq 0$ by ($\mathcal{E}$.3) then inequality (C.15) implies

$$\mathcal{E}(u_\eta(t), t) \leq \mathcal{E}(u_0, 0) + \int_0^t \partial_t \mathcal{E}(u(s), s) ds \leq \mathcal{E}(u_0, 0) + C \int_0^t (1 + \mathcal{E}(u_\eta(s), s)) ds.$$

By ($\mathcal{E}$.3) and Gronwall inequality $-1 \leq \mathcal{E}(u_\eta(t), t) \leq C_1$ for all $t \in [0, T]$ and $\eta > 0$. Then (C.15) implies

$$\int_0^T \mathfrak{f}(u_\eta(t), \dot{u}_\eta(t), t) dt \leq 1 + \mathcal{E}(u_0, 0) + CT(1 + C_1).$$

This concludes the proof. □

Let $\boldsymbol{s}_\eta\colon [0, T] \to [0, S_\eta]$ defined by

$$\boldsymbol{s}_\eta(t) = t + \int_0^t \mathfrak{f}(u_\eta(s), \dot{u}_\eta(s), s) ds \quad \text{and} \quad S_\eta = \boldsymbol{s}_\eta(T). \tag{C.16}$$

Let also

$$\boldsymbol{t}_\eta = \boldsymbol{s}_\eta^{-1}\colon [0, S_\eta] \to [0, T] \quad \text{and} \quad \boldsymbol{u}_\eta = u_\eta(\boldsymbol{t}_\eta)\colon [0, S_\eta] \to V.$$

To write the time-rescaled energy identity we introduce the map $\mathfrak{F}_\eta\colon V\times[0,T]\times V\times(0,\infty)\to[0,\infty)$ defined by

$$\mathfrak{F}_\eta(u,t,v,a)=a\Phi_\eta(v/a)+\Phi_\eta^*(-\partial_u\mathcal{E}(u,t))-a\partial_t\mathcal{E}(u,t)=\mathcal{R}(v)+\mathfrak{E}_\eta(u,t,v,a)-a\partial_t\mathcal{E}(u,t),$$

where $\mathfrak{E}_\eta\colon V\times[0,T]\times V\times(0,\infty)\to[0,\infty)$ is given by

$$\begin{aligned}\mathfrak{E}_\eta(u,t,v,a)&=a\mathcal{V}_\eta(v/a)+a\Phi_\eta^*(-\partial_u\mathcal{E}(u,t))=a\mathcal{V}_\eta(v/a)+\frac{a}{\eta}\min_{\zeta\in V^*}\phi^*(\|\partial_u\mathcal{E}(u,t)+\zeta\|)=\\&=a\mathcal{V}_\eta(v/a)+\frac{a}{\eta}\phi^*\left(\min_{\zeta\in V^*}\|\partial_u\mathcal{E}(u,t)+\zeta\|\right)=a\mathcal{V}_\eta(v/a)+\frac{a}{\eta}\phi^*(\mathfrak{e}(u,t)).\end{aligned}$$

Then (C.12) becomes

$$\mathcal{E}(\boldsymbol{u}_\eta(t),\boldsymbol{t}_\eta(t))+\int_s^t\mathfrak{F}_\eta(\boldsymbol{u}_\eta(r),\boldsymbol{t}_\eta(r),\dot{\boldsymbol{u}}_\eta(r),\dot{\boldsymbol{t}}_\eta(r))dr=\mathcal{E}(\boldsymbol{u}_\eta(s),\boldsymbol{t}_\eta(s))\quad\text{for all } 0\le s<t\le S_\eta.$$

Identity (C.16) implies

$$1=\dot{\boldsymbol{t}}_\eta(s)+\mathfrak{f}(\boldsymbol{u}_\eta(s),\dot{\boldsymbol{u}}_\eta(s),\boldsymbol{t}_\eta(s))\quad\text{for a.e. } s\in(0,S_\eta).\tag{C.17}$$

Now let $\mathfrak{F}\colon V\times[0,T]\times V\times[0,\infty)\to[0,\infty]$ defined by

$$\mathfrak{F}(u,t,v,a)=\mathcal{R}(v)+\mathfrak{E}(u,t,v,a)-a\partial_t\mathcal{E}(u,t)$$

where $\mathfrak{E}\colon V\times[0,T]\times V\times[0,\infty)\to[0,\infty]$

$$\mathfrak{E}(u,t,v,a)=\begin{cases}\iota_K(-\partial_u\mathcal{E}(u,t)) & \text{if } a>0\\ \mathfrak{e}(u,t)\|v\| & \text{if } a=0.\end{cases}$$

**Definition C.3.** We say that a pair $(\boldsymbol{u},\boldsymbol{t})\colon[0,S]\to V\times[0,T]$ is an admissible parametrized curve if

(i) $\boldsymbol{t}$ is non decreasing and absolutely continuous, $\boldsymbol{u}\in\mathrm{AC}_{\mathcal{R}}([0,S],D_E)$ for some $E>0$
(ii) $\boldsymbol{u}$ locally absolutely continuous (with respect to $\|\cdot\|$) in the open set

$$G=\{s\in[0,S]\mid\mathfrak{e}(\boldsymbol{u}(s),\boldsymbol{t}(s))>0\}=\{s\in[0,S]\mid-\partial_u\mathcal{E}(\boldsymbol{u}(s),\boldsymbol{t}(s))\notin K\}$$

(iii) the estimate

$$\int_0^S\mathfrak{f}[\boldsymbol{u},\dot{\boldsymbol{u}},\boldsymbol{t}](s)ds=\int_0^S\mathcal{R}[\dot{\boldsymbol{u}}](s)ds+\int_G\mathfrak{e}(\boldsymbol{u}(s),\boldsymbol{t}(s))\|\dot{\boldsymbol{u}}(s)\|ds<\infty$$

holds.

For every admissible parametrized curve and for a.e. $s\in[0,S]$ is well defined

$$\mathfrak{F}[\boldsymbol{u},\boldsymbol{t},\dot{\boldsymbol{u}},\dot{\boldsymbol{t}}](s)=\begin{cases}\mathcal{R}[\dot{\boldsymbol{u}}](s)+\iota_K(-\partial_u\mathcal{E}(\boldsymbol{u}(s),\boldsymbol{t}(s)))-\dot{\boldsymbol{t}}(s)\partial_t\mathcal{E}(\boldsymbol{u}(t),\boldsymbol{t}(s)) & \text{if } \dot{\boldsymbol{t}}(s)>0\\ \mathcal{R}[\dot{\boldsymbol{u}}](s)+\mathfrak{e}(\boldsymbol{u}(s),\boldsymbol{t}(s))\|\dot{\boldsymbol{u}}(s)\| & \text{if } \dot{\boldsymbol{t}}(s)=0.\end{cases}$$

where is implicit that if $s\notin G$ then $\mathfrak{e}(\boldsymbol{u}(s),\boldsymbol{t}(s))\|\dot{\boldsymbol{u}}(s)\|=0$.

We denote with $\mathscr{A}([0,S],V\times[0,T])$ the set of all admissible parametrized curves and we say that $(\boldsymbol{u},\boldsymbol{t})$ is

(i) non degenerate if $\dot{\boldsymbol{t}}(s)+\mathcal{R}[\dot{\boldsymbol{u}}](s)>0$ for a.e. $s\in[0,S]$
(ii) surjective if $\boldsymbol{t}(0)=0$ and $\boldsymbol{t}(S)=T$
(iii) $\boldsymbol{m}$-normalized with $\boldsymbol{m}\in L^\infty((0,S),(0,\infty))$ if $(\boldsymbol{u},\boldsymbol{t})$ fulfills

$$\dot{\boldsymbol{t}}(s)+\mathcal{R}[\dot{\boldsymbol{u}}](s)+\mathfrak{e}(\boldsymbol{u}(s),\boldsymbol{t}(s))\|\dot{\boldsymbol{u}}(s)\|=\boldsymbol{m}(s)\quad\text{for a.e. } s\in(0,S)$$

**Definition C.4.** A parametrized balanced-viscosity solution is a non degenerate and surjective curve $(\boldsymbol{u},\boldsymbol{t}) \in \mathscr{A}([0,S],V\times[0,T])$ satisfying

$$\mathcal{E}(\boldsymbol{u}(t),\boldsymbol{t}(t)) + \int_s^t \mathfrak{F}[\boldsymbol{u},\boldsymbol{t},\dot{\boldsymbol{u}},\dot{\boldsymbol{t}}](r)dr = \mathcal{E}(\boldsymbol{u}(s),\boldsymbol{t}(s)) \quad \text{for all } 0 \le s < t \le S.$$

**Lemma C.5.** *Let $\boldsymbol{t}_n \in AC([0,S],[0,T])$ non decreasing and surjective. Let $E>0$ and $\boldsymbol{u}_n \in AC([0,S],D_E)$. Assume that there exists $L>0$ such that*

$$\dot{\boldsymbol{t}}_n(s) + \mathfrak{f}(\boldsymbol{u}_n(s),\dot{\boldsymbol{u}}_n(s),\boldsymbol{t}_n(s)) \le L \quad \textit{for a.e. } s \in (0,S) \tag{C.18}$$

*Then up to a extracting a subsequence, $(\boldsymbol{u}_n,\boldsymbol{t}_n)$ converges uniformly on $[0,S]$ to some $(\boldsymbol{u},\boldsymbol{t}) \in \mathscr{A}([0,S],D_E\times[0,T])$. Moreover*

$$\dot{\boldsymbol{t}}(s) + \mathfrak{f}[\boldsymbol{u},\dot{\boldsymbol{u}},\boldsymbol{t}](s) \le L \quad \textit{for a.e. } s \in (0,S)$$

*and the following lower semicontinuity properties holds*

$$\liminf_{n\to\infty} \int_s^t \mathcal{R}(\dot{\boldsymbol{u}}_n(r))dr \ge \int_s^t \mathcal{R}[\dot{\boldsymbol{u}}](r)dr$$
$$\liminf_{n\to\infty} \int_s^t \mathfrak{e}(\boldsymbol{u}_n(r),\boldsymbol{t}_n(r))\|\dot{\boldsymbol{u}}_n(r)\|dr \ge \int_s^t \mathfrak{e}(\boldsymbol{u}(r),\boldsymbol{t}(r))\|\dot{\boldsymbol{u}}(r)\|dr$$
$$\liminf_{n\to\infty} \int_s^t \mathfrak{E}_{\eta_n}(\boldsymbol{u}_n(r),\boldsymbol{t}_n(r),\dot{\boldsymbol{u}}_n(r),\dot{\boldsymbol{t}}_n(r))dr \ge \int_s^t \mathfrak{E}(\boldsymbol{u}(r),\boldsymbol{t}(r),\dot{\boldsymbol{u}}(r),\dot{\boldsymbol{t}}(r))dr$$

*for every $0 \le s < t \le S$ and $\eta_n \in (0,T)$ with $\eta_n \to 0$.*

*Proof. Step 1.* From the fact that $\boldsymbol{t}_n(0)=0$ and that $\dot{\boldsymbol{t}}_n$ are uniformly bounded (by (C.18)), it follows that $\boldsymbol{t}_n$ are uniformly Lipschitz continuous. Then by Ascoli–Arzelà up to extracting a subsequence, $\boldsymbol{t}_n$ converges uniformly to some $\boldsymbol{t} \in C^{0,1}([0,T])$. VERIFICARE CHE $\boldsymbol{t}$ SODDISFA LE PROPRIETÀ RICHIESTE.

*Step 2.* Since $\mathcal{R}$ is continuous and $D_E$ is weakly compact we can prove that for all $\delta>0$ there exists $M_\delta>0$ such that

$$|v-w| \le \delta + M_\delta \mathcal{R}(v-w).$$

Indeed by contradiction assume that there is some $\delta>0$ such that for all $n$ there exist $v_n$ and $w_n$ with

$$|v_n-w_n| \ge \delta + n\mathcal{R}(v-w).$$

Since $D_E$ is weakly compact, we can assume that $v_n \rightharpoonup v$ and $w_n \rightharpoonup w$. Up to extracting a subsequence we can also assume that $v_n \to v$ and $w_n \to w$ in $H$. QUI VA SISTEMATA LA SITUAZIONE. IL MOTIVO È CHE SERVIREBBE CONVERGENZA FORTE NELLA (SEMI)NORMA IN CUI $\mathcal{R}$ È CONTINUA. Then

$$|v-w| \ge \delta + \limsup_{n\to\infty} n\mathcal{R}(v_n-w_n).$$

This is a contradiction because necessarily $v \ne w$ so that $\mathcal{R}(v-w)>0$ and in this case

$$\limsup_{n\to\infty} n\mathcal{R}(v_n-w_n) = \infty.$$

Let $\omega\colon [0,\infty) \to [0,\infty)$ defined by $\omega(r) = \inf_{\delta>0}(\delta + M_\delta r)$. Notice that $\omega$ is non decreasing and $\lim_{r\to 0}\omega(r) = 0$.

*Step 3.* By Jensen inequality and assumption (C.18)

$$\mathcal{R}(\boldsymbol{u}_n(t)-\boldsymbol{u}_n(s)) = |t-s|\mathcal{R}\left(\frac{1}{|t-r|}\int_s^t \dot{\boldsymbol{u}}_n(r)dr\right) \le \int_s^t \mathcal{R}(\dot{\boldsymbol{u}}_n(r))drl \le L|t-s|.$$

Then

$$|\boldsymbol{u}_n(t)-\boldsymbol{u}_n(s)| \le \omega(\mathcal{R}(\boldsymbol{u}_n(t)-\boldsymbol{u}_n(s))) \le \omega(L|t-s|).$$

This implies that $\boldsymbol{u}_n$ are equicontinuous. Then up to extracting a subsequence

$$\lim_{n\to\infty}\sup_{s\in[0,S]}|\boldsymbol{u}(s)-\boldsymbol{u}_n(s)| = 0$$

for some $\boldsymbol{u} \in \mathrm{AC}_{|\cdot|}([0,T],V)$. □

**Theorem C.5** (existence of parametrized solutions)**.** Let $u_\eta \in \mathrm{AC}([0,T],V)$ be solutions to the doubly nonlinear equation (C.3) with fixed initial condition $u_0 \in V$. Consider non decreasing and surjective absolutely continuous time rescalings $\boldsymbol{t}_\eta\colon [0,S] \to [0,T]$ and let $\boldsymbol{u}_\eta = u_\eta(\boldsymbol{t}_\eta)$. Suppose that as $\eta \to 0$

$$\boldsymbol{m}_\eta = \dot{\boldsymbol{t}}_\eta + \mathfrak{f}(\boldsymbol{u}_\eta, \dot{\boldsymbol{u}}_\eta, \boldsymbol{t}_\eta) \overset{*}{\rightharpoonup} \boldsymbol{m} \quad \text{in } L^\infty((0,S)) \tag{C.19}$$

for some $\boldsymbol{m} \in L^\infty((0,S),(0,\infty))$. Then there exists a sequence $\eta_n \to 0$ such that $(\boldsymbol{u}_\eta, \boldsymbol{t}_\eta)$ converges uniformly on $[0,S]$ to a parametrized balanced viscosity solution $(\boldsymbol{u},\boldsymbol{t}) \in \mathscr{A}([0,S], V\times[0,T])$. Moreover $(\boldsymbol{u},\boldsymbol{t})$ is $\boldsymbol{m}$-normalized.

*Proof.* Arguing as in the proof of Lemma C.4 one finds that there exists $E>0$ such that $\boldsymbol{u}_\eta(s) \in D_E$ for all $\eta>0$ and $s\in[0,S]$. Since (C.19) holds, there exists $L>0$ such that $\|\dot{\boldsymbol{t}}_\eta + \mathfrak{f}(\boldsymbol{u}_\eta, \dot{\boldsymbol{u}}_\eta, \boldsymbol{t}_\eta)\|_\infty \le L$. Then the hypotheses of Lemma C.5 are satisfied and we can find a sequence $\eta_n \to 0$ such that $(\boldsymbol{u}_{\eta_n}, \boldsymbol{t}_{\eta_n})$ converges uniformly on $[0,S]$ to some $(\boldsymbol{u},\boldsymbol{t}) \in \mathscr{A}([0,S], V\times[0,T])$. Since $\mathcal{E} \in C^1(V\times[0,T])$ then

$$\mathcal{E}(\boldsymbol{u}_{\eta_n}(s), \boldsymbol{t}_{\eta_n}(s)) \to \mathcal{E}(\boldsymbol{u}(s),\boldsymbol{t}(s)) \quad \text{for all } s\in[0,S]. \tag{C.20}$$

Now, since $\dot{\boldsymbol{t}}_\eta$ are bounded in $L^\infty((0,S))$, up to extracting a subsequence it holds that $\dot{\boldsymbol{t}}_{\eta_n} \overset{*}{\rightharpoonup} \dot{\boldsymbol{t}}$ in $L^\infty((0,T))$. In particular $\dot{\boldsymbol{t}}_{\eta_n} \rightharpoonup \dot{\boldsymbol{t}}$ in $L^1((0,T))$. Then by Lemma C.6

$$\liminf_{n\to\infty}\int_s^t \partial_t\mathcal{E}(\boldsymbol{u}_{\eta_n}(r), \boldsymbol{t}_{\eta_n}(r))dr \ge \int_s^t \partial_t\mathcal{E}(\boldsymbol{u}(r),\boldsymbol{t}(r))dr. \tag{C.21}$$

Now taking the inferior limit as $n\to\infty$ of the energy identity

$$\mathcal{E}(\boldsymbol{u}_{\eta_n}(t), \boldsymbol{t}_{\eta_n}(t)) + \int_s^t \mathfrak{F}_{\eta_n}(\boldsymbol{u}_{\eta_n}(r), \boldsymbol{t}_{\eta_n}(r), \dot{\boldsymbol{u}}_{\eta_n}(r), \dot{\boldsymbol{t}}_{\eta_n}(r))dr \le \mathcal{E}(\boldsymbol{u}_{\eta_n}(s), \boldsymbol{t}_{\eta_n}(s)) \quad \text{for all } 0\le s<t\le T$$

taking into account (C.20), (C.21) and the lower semicontinuity estimates of Lemma C.5

$$\mathcal{E}(\boldsymbol{u}(t),\boldsymbol{t}(t)) + \int_s^t \mathfrak{F}[\boldsymbol{u},\boldsymbol{t},\dot{\boldsymbol{u}},\dot{\boldsymbol{t}}](r)dr \le \mathcal{E}(\boldsymbol{u}(s),\boldsymbol{t}(s)) + \int_s^t \partial_t\mathcal{E}(\boldsymbol{u}(r),\boldsymbol{t}(r))dr$$

for all $0\le s<t\le T$. ADESSO MANCA DA OTTENERE L'ALTRA STIMA (E VOLENDO MOSTRARE CHE SODDISFA LA GIUSTA NORMALIZZAZIONE). □

**Lemma C.6.** *Let $f_n, f, m_n, m\colon (0,T)\to[0,\infty]$ be measurable functions that satisfy*

$$\liminf_{n\to\infty} f_n(s) \ge f(s) \quad \textit{for a.e. } s\in(0,T)$$

$$m_n \rightharpoonup m \quad \textit{in } L^1((0,T)).$$

*Then*

$$\liminf_{n\to\infty} \int_0^T f_n(s)m_n(s)ds \geq \int_0^T f(s)m(s)ds.$$

*Proof.* Let $g_k(s) = \inf_{n\geq k} \min\{f_n(s), n\}$. Notice that $g_k \in L^\infty((0,T))$. Fix $k$ and notice that for all $n \geq k$

$$\int_0^T f_n(s)m_n(s)ds \geq \int_0^T g_k(s)m_n(s)ds.$$

Taking the inferior limit as $n \to \infty$

$$\liminf_{n\to\infty} \int_0^T f_n(s)m_n(s)ds \geq \int_0^T g_k(s)m(s)ds.$$

Now $g_k$ is an increasing sequence and

$$\lim_{k\to\infty} g_k(s) = \lim_{k\to\infty} \inf_{n\geq k} \min\{f_n(s), n\} = \liminf_{n\to\infty} f_n(s) \geq f(s) \quad \text{for a.e. } s \in (0,S).$$

The passing to the limit $k \to \infty$ and using monotone convergence the proof is concluded. □

C.9. **An example.** Let $V = L^2(\Omega)$ with $\Omega \subset \mathbb{R}^d$ open and bounded. Let

$$\mathcal{R}(v) = \int_\Omega |v|dx \quad \text{and} \quad \mathcal{V}(v) = \frac{1}{2}\int_\Omega |v|^2dx.$$

They satisfy the assumptions (D). In particular the continuity of $\mathcal{R}$ follows from Hölder inequality

$$|\mathcal{R}(v) - \mathcal{R}(w)| \leq \int_\Omega ||v| - |w||\, dx \leq \int_\Omega |v-w|dx \leq C\|v-w\|_2.$$

It holds that $K$ is the closed unit ball in $L^\infty(\Omega)$. Indeed if $\xi \in V^* = L^2(\Omega)$ satisfies $\|\xi\|_\infty \leq 1$ then

$$\mathcal{R}^*(\xi) = \sup_{v\in V}\int_\Omega (\xi v - |v|)\, dx \leq \sup_{v\in V}\int_\Omega (\xi v - |v|)\, dx \leq \sup_{v\in V}\int_\Omega (|\xi|-1)|v|dx \leq 0$$

and $\mathcal{R}^*(\xi) \geq 0$ trivially. If $\|\xi\|_\infty > 1$ then there exists $\epsilon > 0$ and $w \in L^1(\Omega)$ such that $\langle \xi, w\rangle \geq 1+\epsilon$. Since $L^2(\Omega)$ is dense in $L^1(\Omega)$ then for all $\delta > 0$ there exist $v \in L^2(\Omega)$ such that $\|w - v\| \leq \delta$. Then for all $\lambda > 0$

$$\mathcal{R}^*(\xi) \geq \langle \xi, \lambda v\rangle - \|\lambda v\|_1 \geq \lambda\left(\langle \xi, w\rangle - \|w\|_1 - \delta(1+\|\xi\|)\right) \geq \lambda\left(\epsilon - \delta(1+\|\xi\|)\right).$$

Choosing $\delta$ such that $\epsilon - \delta(1+\|\xi\|) > 0$ and letting $\lambda \to \infty$ we get $\mathcal{R}(\xi) = \infty$. Since $\mathcal{V}^* = \frac{1}{2}\|\cdot\|_2^2$ and by (C.4)

$$\Phi_\epsilon^*(\xi) = \frac{1}{2\eta}\min_{\zeta\in K}\|\xi-\zeta\|_2^2 = \frac{1}{2\eta}\int_\Omega \min_{|\zeta|\leq 1}|\xi-\zeta|^2dx.$$

By (C.13) it follows

$$\mathfrak{p}(v,\xi) = \|v\|_1 + \|v\|_2 \min_{\zeta\in K}\|\xi-\zeta\|_2.$$

## Appendix D. Energetic solutions

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